Aaron Lorenzo Woods

alwoods@utexas.edu
Website: woods.engineer

Department of Biomedical Engineering
Cockrell School of Engineering
The University of Texas at Austin
Austin, TX 78705

Education

Ph.D. Biomedical Engineering — **Expected Spring 2027**
Doctoral Candidate · Portfolio in Imaging Science · Advisor: Dr. Andrew Dunn — UT Austin
Optical instrumentation and computational methods for chronic neurovascular imaging in stroke recovery

M.S.E. Biomedical Engineering — **July 2024**
The University of Texas at Austin

B.S. Engineering Leadership — **December 2020**
Minor in Commercial Music and Recording — UTEP

Research Experience

Graduate Student Researcher — **2021 – Present**
Department of Biomedical Engineering · Functional Optical Imaging Laboratory — UT Austin
Dr. Andrew Dunn

Built neurovascular imaging systems for stroke studies using a **fiber-laser platform**, **two Spirit-pumped NOPA sources**, and **two-photon/laser speckle contrast imaging** microscopes; developed Python workflows for **SLAVV**, vessel-graph extraction, and registration that supported **Wang-method**, **albumin-mScarlet** vascular-labeling studies, and Dunn lab grant proposals.

Graduate Rotation Student — **Fall 2021**
Department of Biomedical Engineering — UT Austin
Dr. Evan Wang

Conducted **EEG acquisition and analysis** for sleep-related neural studies using repeatable long-duration recording workflows; developed **dry-electrode fabrication SOPs** and bioelectronics prototypes that improved lab documentation and process design.

Graduate Rotation Student — **Summer 2021**
Department of Biomedical Engineering — UT Austin
Dr. Mia Markey

Designed decision-support research for breast cancer patients, examining how emotion, bias, and clinical workflow shaped patient decision-making; developed support-tool concepts around real clinical constraints in collaboration with researchers and clinicians.

Undergraduate Researcher — **2019 – 2020**
Engineering Education · Create Lab — UTEP
Dr. Peter Golding

Developed engineering-education research in advising software, curriculum design, and engineering-business training; built student-facing tools that improved workflows for 200+ students and supported bio-inspired prototyping, course planning, leadership, finance, and professional-development pedagogy.

Undergraduate Researcher — **Summer 2019**
Adult Neurology Ketogenic Diet Therapy Clinic — UW-Madison
Dr. Elizabeth Felton

Analyzed clinical research data from ketogenic-diet epilepsy cohorts using data visualizations to investigate carnitine deficiencies; supported the dataset and analysis pipeline later reported in a Frontiers in Nutrition publication on hypocarnitinemia and seizure control.

Undergraduate Researcher — **Summer 2018**
Biomedical Optics Lab — UW-Madison
Dr. Jeremy Rogers

Engineered a **common-path interferometer** for precise phase-delay measurements using optical-equipment characterization workflows; programmed **automated calibration software** that established repeatable measurement workflows for precision optical experiments.

Publications & Patents

Publications

1. Mihelic SA, Engelmann SA, Sadr M, Jafari CZ, Zhou A, **Woods AL**, Williamson MR, Jones TA, Dunn AK.
"Microvascular plasticity in mouse stroke model recovery: Anatomy statistics, dynamics measured by longitudinal in vivo two-photon angiography, network vectorization." *Journal of Cerebral Blood Flow & Metabolism*. 2024.
doi:10.1177/0271678X241270465

2. Tomar A, Engelmann SA, **Woods AL**, Dunn AK.
"Non-degenerate two-photon imaging of deep rodent cortex using indocyanine green in the water absorption window." *Biomedical Optics Express*. 2024.
doi:10.1364/BOE.520977

3. Chu DY, Ravelli MN, Faltersack KM, **Woods AL**, Almane D, Li Z, Sampene E, Felton EA.
"Hypocarnitinemia and its effect on seizure control in adult patients with intractable epilepsy on the modified Atkins diet." *Frontiers in Nutrition*. 2024;11:1304209.
doi:10.3389/fnut.2023.1304209

4. Engelmann SA, Tomar A, **Woods AL**, Dunn AK.
"Pulse train gating to improve signal generation for in vivo two-photon fluorescence microscopy." *Neurophotonics*. 2023.
doi:10.1117/1.NPh.10.4.045006

5. Niemeier R, Schnoor N, **Woods AL**, Rogers JD.
"Calibration of liquid crystal variable retarders using a common-path interferometer and fit of a closed-form expression for the retardance curve." *Applied Optics*. 2020.
doi:10.1364/AO.408383

Patents

1. Malady B, **Woods AL**, Pogue M, Purani H.
"Theranostic Delivery System." U.S. Provisional Patent Application 63/442436. March 2, 2023.

Honors, Awards & Fellowships

Fellowships

W.M. Keck Foundation Endowed Graduate Fellowship — **2021 – Present**, UT Austin
Thrust Fellowship Supplement — **2021 – 2025**, UT Austin
West Texas Fellowship — **2022 – 2024**, UT Austin
Cockrell School Fellowship — **2024 – 2025**, UT Austin
Ruth L. Kirschstein Imaging Science Fellowship — **2021 – 2022**, NIH
Dean's Fellowship Supplement — **2021**, UT Austin

Honors & Awards

Texas Health Catalyst Finalist — **2023**, Dell Medical Center
Marvin & Ellie Selig Entrepreneurship Award — **2022**, Texas Innovation Center
Imaging Science Best Poster Award — **2022**, UT Austin ISTP
Engineering Leadership Character Award — **2020**, UTEP E-Lead
Hunt Startup Sponsorship — **2020**, Hunt Institute

Teaching Experience

Teaching Assistant **Spring 2025**
Department of Biomedical Engineering · BME 349 Biomedical Instrumentation · Instructor: Dr. Andrew Dunn UT Austin

Taught **BME 349 Biomedical Instrumentation**, held office hours, and guided students through amplifiers, filters, transducers, electrodes, and microcontroller-based instrumentation topics; supported undergraduate learning in core biomedical instrumentation concepts.

Teaching Assistant **Spring 2018**
Department of Engineering Education and Leadership · Design Nature UTEP

Trained students in fabrication-lab safety and Design Nature workspace onboarding, coordinating access to prototyping resources; taught **CAD** and simulation workflows that helped students move from concept to working prototypes.

Leadership & Service

President, Biomedical Optics Graduates Organization (BOGO UT) **2024**
UT Austin · Austin, TX

Led seminars, professional-development programming, and organizational continuity, coordinating with the **SPIE student chapter**; supported outreach including **World Engineering Day** programming for K-12 students.

Board Director, Graduate Student Agency **Feb. 2022 – 2023**
UT Austin · Austin, TX

Supported graduate-student advocacy, professional programming, and community building, coordinating with campus partners; translated student needs into cross-campus initiatives.

Board Member, North American Students of Cooperation (NASCO) **2024 – 2025**
North America

Served on the NASCO board, contributing to governance, strategic planning, and policy development; supported student cooperative housing organizations across North America.

Board Member, ICC Austin **Aug. 2021 – Present**
Cooperative housing nonprofit · Austin, TX

Helped govern a nine-house cooperative housing nonprofit serving about 200 residents through board operations and long-term planning; supported organizational stability.

Industry & Entrepreneurship

Senior Engineering Intern, Thermo Fisher Scientific **2023**
Hillsboro, OR

Automated semiconductor defect-classification imaging workflows with **Python automation** and **TensorFlow/Keras CNN workflows**; applied machine-learning and scientific-imaging methods to large internal datasets in a production software environment.

Co-founder, Therosafe **2020 – Present**
Health-tech venture · Austin, TX

Founded a health-tech venture focused on reducing radiation exposure in theranostic radiopharmaceutical workflows, translating clinical pain points into decision-support tooling, product concepts, and early strategy; co-invented and filed U.S. provisional patent No. 63/442436 and advanced the venture to Texas Health Catalyst finalist status.